Validation and AI use — review package

Decision/user: investment committee; proposed watch/defer. Valuation cutoff September 8, 2026. Build date October 6, 2026. Published snapshot commit and SHA-256 hashes are recorded in evidence/freeze-manifest.json and submission/submission-manifest.csv.

Finance conventions

USD millions and shares millions. Five full forecasts 2027–2031, proportional 2026 operating bridge; business-acquisition cash already spent in H1 excluded from stub. No negative FCFF clipping. FCFF = EBIT - unlevered cash tax + D&A - capex - change in operating NWC - capitalized acquisitions. Purchased research is already deducted in EBIT; no second cash deduction. Tax uses inherited 20% before nondeductible IPR&D, with the exact tax difference used to reconcile equity-side cash. Terminal NOPAT grows at g with reinvestment g/20% ROIC. June cash plus noncurrent investments minus debt; investments book-value/distributable assumption. Operating leases retained in expense; no separate verified pension adjustment. DCF Q2 diluted proxy differs from FY2025 diluted proxies used consistently in approximate peer EV.

Validation register

- Published synthetic fixture independently reproduced: NOPAT 112.5, FCFF 77.5; EV 1,000 + cash 100 - debt 250 - NCI 20 - pension 10 = common equity 820; /100 shares = $8.20.

- Annual base-year mapping: assets 112,476 = liabilities plus equity 112,476; cash 7,268, inventory 13,744. Full mapping in historical-source-map. No cash plug.

- Six annual statement sets: balance sheet, cash roll-forward, CFO/CFI/CFF, PP&E/intangible/debt/equity checks; gaps below $0.01m. FCFF reconciles to CFO + CFI + interest + levered tax - unlevered tax.

- Directional checks: WACC 8% to 10% lowers value; higher bounded growth increases value; terminal growth 2% to 3% increases value at fixed ROIC. Automated tests written before execution by AI; these DO NOT count as the human locked prediction.

- Bridge: EV + cash + investments - debt = equity; positive shares; reverse DCF reprices snapshot within numerical tolerance.

- Highest-risk tests: incoherent perpetuity refused; corrupted cash causes balance/cash checks to fail; out-of-bound assumptions refused. Negative explicit FCFF remains a cash funding cost.

- Peer definitions: positive denominators; FY2025 GAAP EPS/revenue consistently used; annual diluted averages and book balances marked approximate; MRK/PFE differences qualified. Enterprise and equity multiples not averaged.

- Independent definition check: non-GAAP performance margin is not EBIT; direct Q2 primary filing verifies 893.7m Q2 dilution and 7,050 + 47,858 = 54,908 debt.

- Cold run: fresh environment without packages, regenerated model/peer outputs and tests; see output/cold-run.log. HTTP workbench API base/change/error recovery tested separately. Linda still must record her demonstration.

Changed-input verification — AI-assisted execution completed

Linda supplied an independently authored prediction (user attestation) and explicitly requested AI execution. The frozen record and generated result are in evidence/rd-prediction-record.json and output/rd-assisted-test-result.json. Part-2-Reconciliation.md explains actual results and corrected conclusions. The 17% R&D case is outside the ordinary 18%–24% range: normal refusal is logged; an isolated exploratory extension leaves production safeguards unchanged. Value rises from

$754.19 to $810.23 (+7.43%), below the predicted $820–$880 interval. Earlier supplied $851.07 is superseded. Watch/defer remains the supplied stance. No human AI-off execution is claimed; the separate locked_test.py guard is unchanged. This does not establish compliance with an AI-off requirement if applicable.

Named AI use

| Tool / exposed model | Date | Material contribution | Check | Disposition |

|---|---|---|---|---|

| Codex / GPT-6, Work mode; finer build version not exposed | October 6, 2026 | Integrate statements with FCFF, timing, interface, tests, peer EV/revenue, prose/scripts | Primary LLY/MRK/PFE filings; manual synthetic arithmetic; accounting identities; clean run | Accept/modify as confirmed in Student-Review.md; human locked test remains incomplete |

| Prior Codex assistance; exact model/version not recorded in old README | Prior labs, actual interaction dates to confirm | Standalone DCF, statements, P/E and sensitivity documentation | Existing source trails and rerun outputs; no invented version/date | Qualified as inherited work |

| Prior Lab 4 short-term investment claim ($60m) | Reviewed October 6 | Extra asset in EV bridge | Q2 balance sheet shows only cash/current assets and 3,856 noncurrent investments; no separate line | Rejected as unsupported addition; integrated bridge omits it |

The independent evidence proves particular inputs/definitions, not all judgments. Passing AI-authored tests is not proof that the economic forecast is true. Linda remains responsible for sources, personal confirmations, originality boundary and explaining the system.

Limits and stop rules

Terminal EV share 78.4%; WACC scenario rather than independently estimated CAPM. September 8 market closes were independently rechecked October 6; Linda confirmed the corrected assumption review on October 6; unresolved assumptions remain qualified. Product demand, pricing and capacity lack a separate causal patient model. Spending cuts do not alter modeled clinical/capacity success. Fixed margins include D&A so capex does not directly change modeled EBIT. Simplified acquisitions, fixed tax/other claims, liquidity backstop, year-end/proportional timing, average-share proxies and nonoperating investments constrain accuracy. Stop issuing a valuation if accounting, funding, domain or perpetuity checks fail; reconsider the action when downside evidence is corroborated.

Executed range-ranking comparison - October 19

The reproducible range_ranking.py compares 2031 operating profit, FCFF and per-share output spans across three bounded sets. Growth ranks first over the original ranges; narrowing growth to +/-0.5 percentage points makes R&D first for all three metrics. FCFF ranks R&D, capex, growth, inventory in that narrower set; value ranks R&D, growth, capex, inventory. This is range-dependent sensitivity, not a new forecast or statistical confidence interval. Base assumptions and recommendation are unchanged. See docs/Range-Ranking.md and output/range-ranking.json. Fresh-environment verification and all 11 tests passed again. A student-recorded cold run and the independent human exercise remain unverified/incomplete.